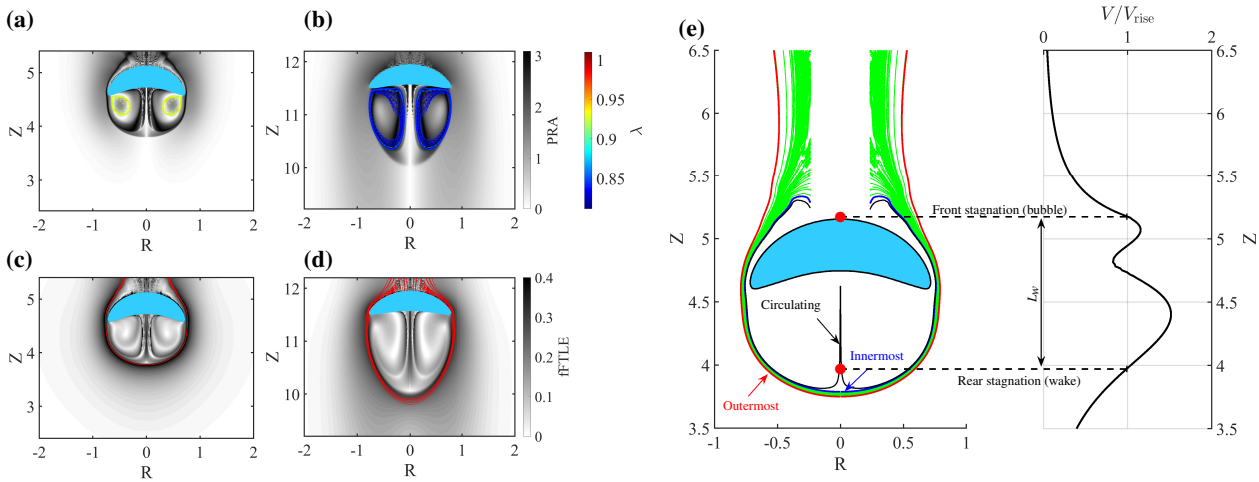

Figure 2: Primary wake of a rising bubble visualized using PRA fields for case C_6 at , highlighting regions of rotational coherence represented by elliptic LCSs, color-coded with the stretching factor λ. The solid line denotes the outermost elliptic LCS, while the dashed lines indicate the inner ones. Corresponding forward FTLE at fields illustrating material transport barriers in the wake, shown as red solid lines. (c) Enlarged Length of the primary wake measured from the outermost hyperbolic LCS and compared with the experimental correlation of **?**].